

LGBT Center Library Catalog

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Problem Statement / Specifications / Requirements

The CWRU LGBT Center hosts a small library containing 4 shelves of books, DVDs and other borrowable documents. The current Library Management System is a Google Sheet containing most of the existing catalog, and a Google Form for patrons to check out and return books. This current system has significant overhead for both patrons who struggle to find and checkout relevant materials as well as LGBT Center employees who are required to meticulously update the spreadsheet and personally track patrons' due dates.

Solutions

We intend to develop a full-stack web application and database that provides a searchable interface for patrons to browse relevant material, checkout, renew, and track their due dates. The application should also contain accessible Library Management features for LGBT Center Staff to update, add and remove items without having technical knowledge of the backend. We hope to utilize university resources to host the application on dedicated server space with a public-facing website for future LGBT Center staff and visitors to use.

The application interface should be easy for the average reasonable person to use, secure to protect the privacy of patrons, and appealing to look at. The database structure should be robust enough to need minimal maintenance in the future, to minimize the burden on the LGBT Center staff. We will nonetheless provide documentation for future updates.

Background & Related Work

Fei has worked on similar Library Management backend projects for courses such as CSDS293—Java-based Library Management software architecture, and CSDS341—MySQL database and rudimentary JDBC front end.

We are drawing inspiration from existing public library catalogs such as the Heights Libraries (<https://heightslibrary.org>), the Cuyahoga County Public Library (<https://cuyahogalibrary.org/>), the King County Library System (<https://kcls.org/>), and the Cleveland Public Library (<https://cpl.org>).

There are a few full stack libraries which either we have experience in or the school uses, which we are considering for the implementation of our project, including

MERN (mongoDB , express.js, [react.js](#), [node.js](#)), LAMP (linux, apache, mysql, php), and Drupal. Additionally, we can experiment with a Spring Boot back-end to prioritize security.

Finally, we will be utilizing the university's CAS Single Sign-On system in order to manage logging into and out of the library catalog, documentation for which can be found at

<https://case.edu/utech/help/knowledge-base/cwru-network-id-passphrase/cas-single-sign-cwru-kba>.

Preexisting database schema:

As stated above, the current method used to keep track of the library catalog is a spreadsheet which can be browsed by potential clients and must be manually updated by staff members, which can be found at

https://docs.google.com/spreadsheets/d/1tbhRvgI_Dn5JIICZVFXtj26xtTF0JUEa-iRT5JDvmvs/htmlview. Patrons can check out books by filling out the Google form linked in the spreadsheet, at <https://forms.gle/pk3VKTN5Bux59rTp9>.

Work to be done (Timeline)

1. Contact UTech to negotiate for server space
2. Outline poster
3. Set up a demonstrative system with barebones:
 - Database with sample data
 - Catalog
 - ISBN number
 - searchable info (author, title, date published, format content tags)
 - Total copies + copies available
 - Users
 - caseID
 - Name (for display)
 - Level of auth (use eNUM?)
 - Checkouts (linking table) – delete entry when book is checked back in, to preserve privacy.
 - caseID
 - ISBN
 - Date checked out
 - Javascript web interface with
 - **Search** that **Reads** catalog entries
 - Account function: 3 levels of authentication

- Manager auth: **update** user authorizations, **update** checkout restrictions (max checkouts, duration of checkouts, max number of renewals)
 - Employee auth: **update** check-outs, **update/create/remove** catalog items
 - Patron auth: **read** catalog entries, **write** checkouts
4. Update poster, update documentation
 5. Iteratively improve this system, work with UTech
 - Verify login w/ CASE SSO
 - Webservices:
 - Email confirmation upon checking out a book
 - Email notification for overdue books
 - Submission form for suggesting new books, which emails to the Managers and Employees
 - SANITIZING Inputs to prevent SQL injection
 6. Make it look good
 - Use design kits (Tailwind CSS/Bootstrap/etc..) with a front-end specific framework (React/Angular/Vue)

Throughout work, we will update the poster.

Management Plan & Accountability

This project can be divided into 2 teams: database backend and web interface front end. The role of the frontend team will be to create the look and feel of the website using HTML, CSS, and JavaScript. The backend team will be responsible for managing the creation of the database, as well as managing database queries sent from the frontend (including sanitizing inputs as necessary). Additionally, they will have to handle users and authentication through the CAS system. Someone in the group will also need to contact UTech to discuss hosting the site on the school's servers.

In preparation for the demo on October 17, we should have a simple version of the database setup with sample entries, as well as a draft of the website design, and a way for the website to communicate with the database to simulate checking out, renewing, and checking in books. Sample users will be utilized without needing to be authenticated with the CAS system for the purpose of the demo.

In between the demo and the end of the semester, we will integrate the CAS system into the account system and clean up the look and feel of the website. We will also have the website hosted on a server by that point so that it can be used by anyone wishing to check books out of the LGBT Center library.

In order to keep on track with this plan, we will keep in regular contact over a group chat and aim to have at least one in-person meeting per week, where we can evaluate the plan moving forward and address any issues which have arisen since the last meeting. We will also keep track of who accomplished what with a spreadsheet:

 Team Planning Spreadsheet .